

Question

The organic extractant HA can extract Ce^{3+} from an aqueous solution to the organic phase. HA is almost insoluble in water, but exists in the organic phase only as a dimer of H_2A_2 . To determine the specific form of this cerium complex in the organic phase, the following experiment was performed:

A tracer amount of $^{144}\text{Ce}(\text{NO}_3)_3$ was dissolved in water and acidified with a small amount of nitric acid to prepare an aqueous phase containing $^{144}\text{Ce}^{3+}$. 2.50 mL of the above aqueous phase and 2.50 mL of the organic phase containing HA were taken and mixed thoroughly for extraction at room temperature. After equilibrium was reached, 1.00 mL each of the aqueous phase and the organic phase were taken, and the ratio of their radioactivity intensities was measured. Another portion of the equilibrated aqueous phase was taken and its pH value was measured. When the initial concentration of H_2A_2 is $c_0(\text{H}_2\text{A}_2) = 0.240 \text{ mol/L}$, the ratio of radioactivity intensity of the organic phase to the aqueous phase at equilibrium was measured to be 2929:416, and the aqueous phase $[\text{H}^+] = 3.39 \times 10^{-2} \text{ mol/L}$. When $c_0(\text{H}_2\text{A}_2) = 0.120 \text{ mol/L}$, the ratio of radioactivity intensity of the organic phase to the aqueous phase at equilibrium was measured to be 1542:1682, and the aqueous phase $[\text{H}^+] = 3.42 \times 10^{-2} \text{ mol/L}$. It is known that the organic phase does not contain nitrate ions.

Please determine the equilibrium constant of the extraction reaction through calculation and reasoning, and select the correct equilibrium constant from the following options:

(Hydrolysis of Ce^{3+} and complexation of Ce^{3+} with NO_3^- are not considered. A calculation error within 5% is considered correct, and take two significant figures).

A. All other options are incorrect

B. 0.010

C. 0.014

D. 0.025

E. 0.30

F. 1.5

G. 0.52

H. 0.030

I. 0.0022

Answer

Correct Answer: **A**

Detailed Explanation

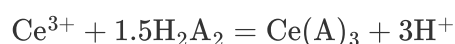
The extracted substance is electrically neutral because there are no nitrate radicals, and Ce^{3+} is coordinated with at least three A^- ions.

CHECKPOINT

1 PTS

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Consider the possible extraction reaction equation:



From the two extraction experiments, the ratio of radiation intensities is the ratio of $\text{Ce}(\text{III})$ concentrations, so the equilibrium constant can be listed:

$$K = \frac{[\text{H}^+]^3[\text{Ce}(\text{A})_3]}{[\text{H}_2\text{A}_2]^{1.5}[\text{Ce}^{3+}]}$$

CHECKPOINT

1 PTS

$$K = \frac{[\text{H}^+]^3[\text{Ce}(\text{A})_3]}{[\text{H}_2\text{A}_2]^{1.5}[\text{Ce}^{3+}]}$$

Since $\text{Ce}(\text{III})$ is a tracer amount, $c_0(\text{H}_2\text{A}_2) \approx [\text{H}_2\text{A}_2]$.

CHECKPOINT

1 PTS

Since Ce(III) is a tracer amount, $c_0(\text{H}_2\text{A}_2) \approx [\text{H}_2\text{A}_2]$

Therefore, the following equation can be listed according to the two experiments:

$$K = \frac{0.0339^3 \times 2929}{416 \times 0.24^{1.5}} = \frac{0.0342^3 \times 1542}{1682 \times 0.12^{1.5}}$$

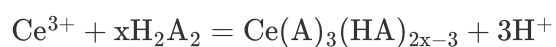
Obviously, this equation does not hold. Therefore, some HA is also coordinated with Ce(III) in the organic phase;

CHECKPOINT

1 PTS

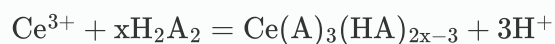
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At this time, the extraction reaction can be written as



CHECKPOINT

1 PTS



List the equation according to the equilibrium constant:

$$K = \frac{0.0339^3 \times 2929}{416 \times 0.24^x} = \frac{0.0342^3 \times 1542}{1682 \times 0.12^x}$$

Solving for $x = 2.9 \approx 3$. That is, the existence form of Ce(III) in the organic phase is $\text{Ce(A)}_3(\text{HA})_3$.

CHECKPOINT

1 PTS

The existence form of Ce(III) in the organic phase is $\text{Ce(A)}_3(\text{HA})_3$

Substituting, we get the equilibrium constant $K = 0.019$.

CHECKPOINT

1 PTS

Equilibrium constant $K = 0.019$

Therefore, all options are incorrect. Option A is correct.